



## Pyrolysis of municipal plastic wastes II: Influence of raw material composition under catalytic conditions

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### ABSTRACT

In this work, the results obtained in catalytic pyrolysis of three plastic waste streams which are the rejects of an industrial packing wastes sorting plant are presented. The samples have been pyrolysed in a 3.5 dm<sup>3</sup> reactor under semi-batch conditions at 440 °C for 30 min in nitrogen atmosphere. Commercial ZSM-5 zeolite has been used as catalyst in liquid phase contact. In every case, high HHV gases and liquids which can be useful as fuels or source of chemicals are obtained. A solid fraction composed of the inorganic material contained in the raw materials and some char formed in the pyrolysis process is also obtained. The zeolite has shown to be very effective to produce liquids with great aromatics content and C3–C4 fraction rich gases, even though the raw material was mainly composed of polyolefins. The characteristics of the pyrolysis products as well as the effect of the catalyst vary depending on the composition of the raw material. When paper rich samples are pyrolysed, ZSM-5 zeolite increases water production and reduces CO and CO<sub>2</sub> generation. If stepwise pyrolysis is applied to such sample, the aqueous liquid phase can be separated from the organic liquid fraction in a first low temperature step.

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### 1. Introduction

Packaging waste management is at present a key concern in Europe. The growing waste generation has forced the European Union (EU) to face the problem, taking specific policies which have tried to boost sustainable development since the 1970s (Alwaeli, 2010). The Council Directive 94/62/EC established for the first time the levels to be recycled concerning packaging wastes and these levels were re-established and hardened by the Council Directive 2004/12/EC, which amended the former. Although this Directive aims at reducing the production of packaging waste, a general increase in per capita quantities of packaging being put on the European market was reported in the period 1997–2007, which lead to the corresponding increase in packaging waste generation in the EU-27 Member States (European Environment Agency, 2010).

There are large differences in the management of packaging wastes among Member States, depending mainly on the use and generation of these wastes. In Spain, packaging wastes are collected in specific municipal containers and then sent to industrial separation and classification plants, where the wastes are separated in different fractions (PE, PET, mixed plastics fraction, steel, tetra-brick, aluminium cans, etc.) for their subsequent mechanical recycling. In such plants, a significant amount of the income mate-

rials cannot be properly classified or separated and it is rejected. Although it is mainly composed of plastics, many different intermingled packaging materials are found in this rejected fraction and therefore it cannot be mechanically recycled; for this reason, the rejected fraction is nowadays incinerated or landfilled. The average rejected percentage of the income materials in Spanish separation plants is around 45 wt.% (Ecoembalajes España, 2009).

Pyrolysis could be an alternative to landfill or incineration for these rejected streams. In the pyrolysis process (heating in an oxygen free atmosphere), the organic components of the material are decomposed generating liquid and gaseous products, which can be useful as fuels and/or sources of chemicals. The inorganic ingredients (fillers, metals, etc.) remain practically unaltered and free of the binding organic matter; therefore metals could be separated, and the remaining solid (non-metallic inorganic compounds + char) may be reused as additive, filler for plastics, pigment, etc., although its potential applications strongly depend on the specific composition of the raw material which directly conditions the pyrolysis solids composition. Anyhow, as a last resort, it would be a minimum waste to be landfilled.

Although a lot of papers have been published in the last years concerning thermal and catalytic decomposition of plastics (e.g. Kaminsky et al., 1976, 2004; Kaminsky and Franck, 1991; Williams and Williams, 1996; Williams and Badri, 2004; Serrano et al., 1998, 2009), it is difficult to find information in the literature about pyrolysis of real fractions of plastic wastes coming from packaging applications (Kaminsky et al., 1997; Kaminsky and Kim, 1999),

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### Nomenclature

|      |                           |     |                              |
|------|---------------------------|-----|------------------------------|
| HDPE | high density polyethylene | PS  | polystyrene                  |
| PE   | polyethylene              | PET | poly(ethylene terephthalate) |
| PP   | polypropylene             | PVC | poly(vinyl chloride)         |

**Table 1**

Composition of the samples pyrolysed (wt.%).

|                           | Material   | Sample 1 (film rich sample) | Sample 2 (paper rich sample) | Sample 3 (glass rich sample) |
|---------------------------|------------|-----------------------------|------------------------------|------------------------------|
| Packaging plastics        | HDPE       | 13.44                       | 5.26                         | 18.77                        |
|                           | PP         | 9.63                        | 8.19                         | 21.94                        |
|                           | PS         | 6.60                        | 9.79                         | 11.38                        |
|                           | PET        | 2.88                        | 7.33                         | 3.23                         |
|                           | PVC        | 4.28                        | 1.42                         | 0.79                         |
|                           | PE film    | 50.55                       | 9.40                         | 2.14                         |
|                           | PP film    | 4.92                        | 2.07                         | 1.73                         |
|                           | Subtotal   | 92.30                       | 43.46                        | 59.98                        |
| Other packaging materials | Tetra-brik | 2.73                        | 7.82                         | –                            |
|                           | Others     | 1.90                        | 7.08                         | 5.34                         |
|                           | Subtotal   | 4.63                        | 14.90                        | 5.34                         |
| Inappropriate materials   | Paper      | 2.80                        | 33.25                        | 4.19                         |
|                           | Wood       | 0.07                        | 0.28                         | 3.32                         |
|                           | Glass      | 0.01                        | 0.60                         | 16.18                        |
|                           | Inerts     | –                           | 1.04                         | 1.78                         |
|                           | Others     | 0.21                        | 6.49                         | 9.21                         |
|                           | Subtotal   | 3.07                        | 41.66                        | 34.68                        |

and even more difficult papers concerning catalytic degradation, since catalysts can be deactivated by the presence of inert materials and heteroatoms (Cl, N, S) which form usually part of these type of wastes (Ding et al., 1997; Aguado and Serrano, 1999).

In a previous work (López et al., 2010) the authors studied the variability and behaviour in thermal pyrolysis at 500 °C of the rejected stream of a packaging classification and separation plant. The authors proved that the composition of the rejects of a specific industrial waste separation plant may significantly vary in the course of time and this variability has a strong influence in the pyrolysis process and in the products obtained. The objective of this paper, which is the second part of the author's previous work, is to study the catalytic degradation of these real samples in presence of ZSM-5 zeolite, an acid solid catalyst which has been reported to be especially appropriate in the cracking of plastic feedstocks (e.g. Serrano et al., 2000; Miskolczi et al., 2004; Aguado et al., 2007). The influence of the catalyst in the process and products quality as a function of the composition of the raw material will be discussed, and some results will be compared with those obtained in the first part of the work (thermal pyrolysis) in order to better understand the possibilities of the pyrolysis process. The discussion will be supported with the complete characterisation of the pyrolysis products.

## 2. Experimental

### 2.1. Samples and catalyst

Three real samples coming from the rejects of a packaging separation and classification industrial plant located in Amorebieta, in the north of Spain, have been pyrolysed. This industrial plant sorts the packaging wastes of most of the province of Biscay. Informa-

tion about the process of the separation plant has been given elsewhere, together with the complete characterisation of the samples and the discussion about their composition and characterisation (López et al., 2010). In order to provide enough information to follow this paper, a summary of the main components as well as the characterisation results of the samples are presented again in this paper in Tables 1 and 2.

The catalyst used in the experiments was a commercial ZSM-5 zeolite provided by Zeolyst International; its main characteristics are shown in Table 3. In all the experiments it was used as was received in a plastic/catalyst ratio of 10/1 and in liquid phase contact mode. In this catalytic configuration the catalyst is mixed with the sample in the pyrolysis reactor; liquid phase contact mainly differs from vapour phase contact (in this configuration the catalyst is placed in a secondary reactor where pyrolysis vapours are

**Table 2**

Moisture, ash and elemental composition (wt.%) and HHV (MJ kg<sup>-1</sup>) of the samples pyrolysed.

| Sample                     | Film rich sample   | Paper rich sample                                       | Glass rich sample                    |
|----------------------------|--------------------|---------------------------------------------------------|--------------------------------------|
| Moisture                   | 0.6                | 2.0                                                     | 1.0                                  |
| Ash                        | 2.4                | 11.4                                                    | 28.2                                 |
| C                          | 81.0               | 61.1                                                    | 58.7                                 |
| H                          | 13.7               | 8.9                                                     | 8.7                                  |
| N                          | 0.1                | 0.3                                                     | 0.5                                  |
| S                          | <0.1               | <0.1                                                    | <0.1                                 |
| Cl                         | 0.7                | 2.4                                                     | 1.0                                  |
| Others <sup>a</sup>        | 1.4                | 13.8                                                    | 1.8                                  |
| H/C ratio                  | 2.0                | 1.7                                                     | 1.8                                  |
| Gross formula <sup>b</sup> | CH <sub>2.02</sub> | CH <sub>1.75</sub> Cl <sub>0.01</sub> O <sub>0.17</sub> | CH <sub>1.78</sub> O <sub>0.02</sub> |
| HHV                        | 41.4               | 29.4                                                    | 30.6                                 |

<sup>a</sup> By difference.

<sup>b</sup> Considering others = oxygen.

**Table 3**

Textural and acid properties of the ZSM-5 zeolite.

|                                                            |       |
|------------------------------------------------------------|-------|
| BET surface area ( $\text{m}^2 \text{g}^{-1}$ )            | 412.0 |
| External surface area (ESA) ( $\text{m}^2 \text{g}^{-1}$ ) | 65.9  |
| Micropore volume (MPV) ( $\text{cm}^3 \text{g}^{-1}$ )     | 0.1   |
| Total pore volume (TPV) ( $\text{cm}^3 \text{g}^{-1}$ )    | 0.4   |
| Total acidity ( $\text{mmol NH}_3 \text{g}^{-1}$ )         | 0.2   |

reformed) in the fact that the catalyst directly influences in the pyrolysis reactions. Although sometimes the consumption of catalyst is higher in liquid phase contact, this configuration was used in this work with the aim of studying the effect of catalyst in the pyrolysis reactions.

## 2.2. Thermogravimetric analyses and pyrolysis experiments

The thermal behaviour of the samples was studied using a Mettler Toledo TGA/SDTA851 analyser. The analyses were conducted with 7.5 mg samples, which were heated under nitrogen flow ( $50 \text{ mL min}^{-1}$ ) to  $600^\circ\text{C}$  at a rate of  $20^\circ\text{C min}^{-1}$ . The temperatures of the maximum degradation rates were determined from the derivative thermogravimetric (DTG) plot. The thermogravimetric analyses were carried out without catalyst since they were used to obtain analytical information.

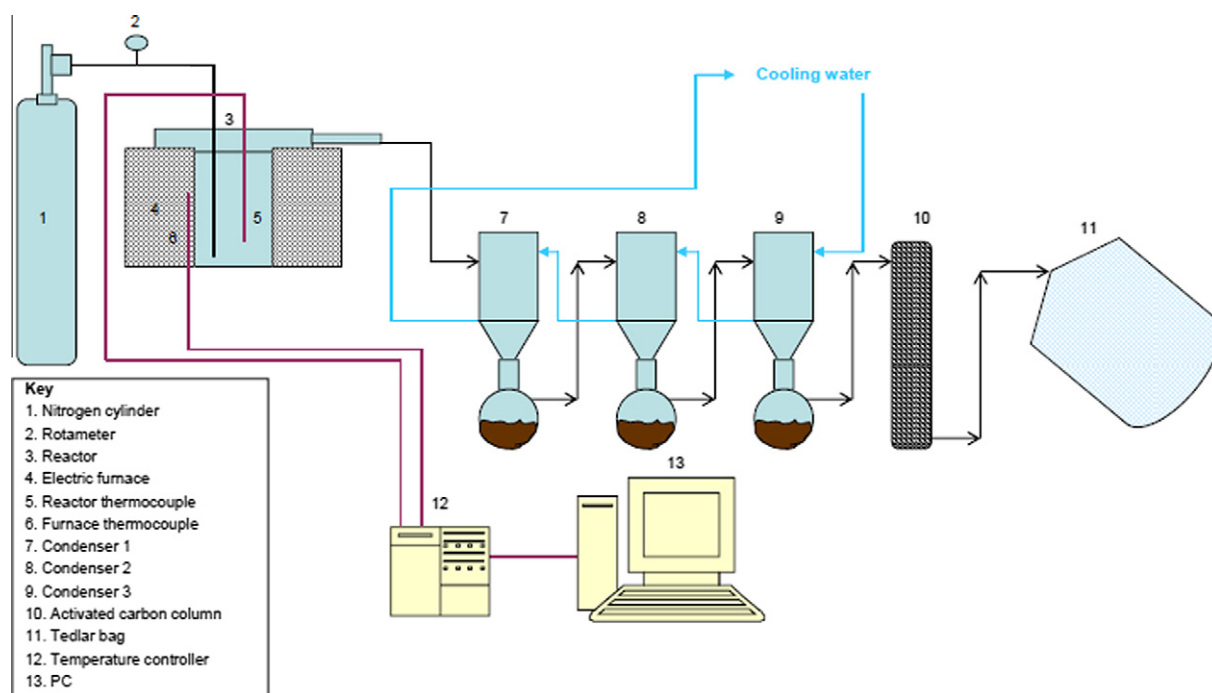
The pyrolysis experiments were carried out at  $440^\circ\text{C}$  in nitrogen atmosphere, using an unstirred stainless steel  $3.5 \text{ dm}^3$  reactor in semi-batch operation. In a typical run, 100 g of sample with a particle size of 8 mm are mixed with 10 g of catalyst and placed into the reactor. Nitrogen is passed through at a rate of  $1 \text{ dm}^3 \text{ min}^{-1}$  and the system is heated at a rate of  $20^\circ\text{C min}^{-1}$  to  $440^\circ\text{C}$ , and maintained there for 30 min. The final temperature ( $440^\circ\text{C}$ ) of the process but pyrolysis begins below that temperature; in the catalytic experiments, the first pyrolysis vapours can be seen between  $270$  and  $300^\circ\text{C}$ . The parameters used in this paper ( $440^\circ\text{C}$ , 30 min) are optimised operating conditions as far as energy efficiency, conversion and liquid yields is concerned. These

optimised conditions were established by the authors after specific research concerning this topic.

During each run the vapours leaving the reactor flow to a series of running water cooled gas–liquid separators where the condensed liquids are collected. The uncondensed products are passed through an activated carbon column and collected as a whole in Tedlar plastic bags. Some tests carried out by the authors with and without the activated carbon column (de Marco et al., 2009) showed that such column retained all the aromatic products as well as some liquid products which otherwise passed to the Tedlar bag. A rather small proportion of C4–C6 compounds were also retained, but there was not a significant difference between the composition and HHV of the pyrolysis gases obtained with and without the activated carbon column.

The flow sheet of the experimental setup is presented in Fig. 1. As it can be seen in the figure, the thermocouple which measures and controls the process is placed in the middle of the reactor chamber. This means that although the reactor is an unstirred one and plastics have low thermal conductivity, it is guaranteed that the whole plastic sample reaches at least the preset temperature. On the other hand and concerning  $\text{N}_2$  carrier gas distribution, there is a diffusion plate inside the reaction chamber (in the bottom of the reactor) which distributes nitrogen all around the reaction chamber.

The amount of solids (products left in the reactor) and liquids obtained are weighed and the pyrolysis yields calculated as weight percentage with respect to the raw material pyrolysed (100 g). The amount of input zeolite is subtracted from the total amount of solids and the yield is calculated with respect to input plastic material (without catalyst). Gas yields are normally determined by difference. In some experiments specifically devoted to directly quantifying the amount of gases, a closure of the mass balance of about 90 wt.% was obtained. In such experiments, total gas quantification was made as follows: (1) mass quantification ( $\text{mg mL}^{-1}$ ) of each gaseous component was determined by GC–FID/TCD; (2) total pyrolysis gas volume was measured and (3) the gaseous compounds quantities were extrapolated to the total volume.

**Fig. 1.** Flow sheet of the experimental set-up used.

The results of the pyrolysis yields which are presented in Section 3 of this paper are the mean value of at least three pyrolysis runs carried out in the same conditions and which did not differ more than three points in the percentage.

### 2.3. Analytical techniques

Samples of about 25 kg were collected at the industrial sorting plant. Then, representative samples of 6 kg were selected by cone and quartering procedure and those samples were used for the identification and quantification of components. Visual inspection as well as simple identification techniques (flame colour, density, and in doubtful cases FT-IR analysis) were used for the composition determination. Once the components were identified and quantified, the samples were ground using a cutting mill suitable for plastics with a sieve grate of 8 mm, which generates a particle size small enough ( $\leq 8$  mm) to be able to extract homogeneous 100 g samples for the experiments. Additionally, 100 g of the sample with a particle size  $\leq 8$  mm were more finely ground ( $\leq 1$  mm) for characterisation purposes under cryogenic conditions using a small cutting mill.

Both the raw materials and the solid and liquid pyrolysis products obtained were thoroughly characterised using the following analytical techniques. The moisture and ash contents of the samples were determined by thermogravimetric analysis, according to D3173-85 and D3174-82 ASTM standards, respectively, and the elemental composition with automatic CHN and S analysers. Method 5050 of the Environmental Protection Agency (EPA) of the United States was used for Cl determination. The higher heating value (HHV) was determined with an automatic calorimetric bomb.

Additionally, pyrolysis liquids were also analysed by gas chromatography coupled with mass spectrometry detector (GC-MS). When the match quality of the identification result provided by the MS search engine was lower than 85%, the result was not considered valid and these compounds are classified as “Not identified” in the corresponding tables in this paper. The compound names correspond to the tentative assignments provided by the MS search engine, contrasted as far as possible with bibliographic data and sometimes with standards. Concerning pyrolysis gases, they were analysed by means of a gas chromatograph coupled with a thermal conductivity and a flame ionization detectors (GC-TCD/FID). The HHV of the gases was theoretically calculated according to their composition and to the HHV of the individual components.

The textural properties of the catalyst were determined by means of nitrogen adsorption-desorption isotherms at 77 K in AUTOSORB-1 Quantachrome equipment. Surface areas were calculated by means of BET equation and external surface areas were obtained applying the *t*-plot method. Total pore volume was measured at  $P/P_0 = 0.99$ . Acidity of the catalysts was measured by temperature-programmed desorption (TPD) of ammonia on a Micromeritics AutoChem 2910 instrument.

## 3. Results and discussion

### 3.1. Thermogravimetric analyses

For a better understanding of the thermal behaviour of the samples, their TGA plots have been included in Fig. 2; Fig. 2a shows the weight loss of the samples as a function of temperature and Fig. 2b the first derivative of the weight loss. It can be seen in Fig. 2a that all of them reach their final weight loss at temperatures over 500 °C, being this weight loss different from one sample to another due to the differences in composition shown in Table 1. In general terms, the lower proportions of inorganic and char forming mate-

rials in the film rich sample make this sample have the highest weight loss ( $\approx 95$  wt.%). On the contrary, the paper and glass rich samples reach lower weight loss, i.e., 80 and 70 wt.%, respectively.

Fig. 2b gives more information about the composition of the samples. The paper rich sample shows a quite higher peak near 500 °C than the other samples, which indicates that this sample contains higher quantities of polyolefins than the other ones, which is in good agreement with the data reported in Table 1. On the contrary, the paper and glass rich samples shows a more noticeable peak at around 440 °C than the film rich sample, which is indicative of the higher amount of PS and PET in the composition of the formers. Additionally, a small peak around 300 °C can be seen in all the samples. This corresponds to PVC first decomposition but also to paper decomposition; this is the reason this peak is the highest one for the paper rich sample.

### 3.2. Pyrolysis yields

The pyrolysis yields obtained with the three samples tested are presented in Table 4. It can be seen that the yields highly vary depending on the sample pyrolysed. The film rich sample yields 49.9 wt.% of gases and 41.5 wt.% of organic liquids, which means that in spite of the low temperature used (440 °C), more than 90 wt.% of the plastic waste has been converted. This is an interesting result since 65 wt.% of this sample is PE, and it has been reported that the maximum thermal degradation rate of PE is achieved between 470 and 480 °C (Serrano et al., 2005). In the previous paper of the authors (López et al., 2010) very similar conversion (92.2 wt.%) was obtained with the very same sample at 500 °C without catalyst. Therefore, pyrolysis can be carried out at lower temperatures than thermal runs when ZSM-5 zeolite is used as catalyst, obtaining quite similar results. The particular effect of this catalyst has also been reported by other authors (Aguado et al., 2007).

The paper rich sample produced lower quantities of organic liquids (25.5 wt.%) and gases (29.4 wt.%) than the film rich sample and it generated an aqueous phase in the liquid fraction and the highest char yield (16.9 wt.%). These effects were also observed in non-catalytic pyrolysis in the previous work published by the authors (López et al., 2010) and they are attributed to the high content in cellulose based materials of this sample. The aqueous liquid fraction obtained in this paper (17.1 wt.%) is higher than that obtained then (12.4 wt.%), due to the greater water generation which is produced in presence of acid solid catalysts during cellulose based materials pyrolysis (Aho et al., 2007; Pan et al., 2010). This also explains the aqueous phase generation in the pyrolysis of the glass rich sample (3.3 wt.%), which did not yield aqueous fraction in the thermal experiments but which also contains a relatively high proportion of cellulose based materials (paper + wood > 7 wt.%).

It is also worth noting that the glass rich sample produced quite a high char yield (12.9 wt.%) compared to that obtained in the previous paper of the authors (5.3 wt.%). So the zeolite strongly promotes char formation with this sample, which is in agreement with what has been reported by Serrano et al. (2000) and Williams and Badri (2004), that zeolitic catalysts contribute to char formation. With the other samples, however, char production is only slightly higher than in the thermal experiments (6.2 wt.% now and 5.4 wt.% in the previous work with the film rich sample; 16.9 wt.% now and 15.2 wt.% then with the paper rich sample). So it seems that some kind of interaction among the materials of the glass rich sample and the catalyst itself takes place. It may be suggested that the great quantities of nonvolatile components (glass, other inert materials and catalyst) which are mixed with the polymeric materials in such sample contribute to char formation just because there is some physic or steric hindrance in the

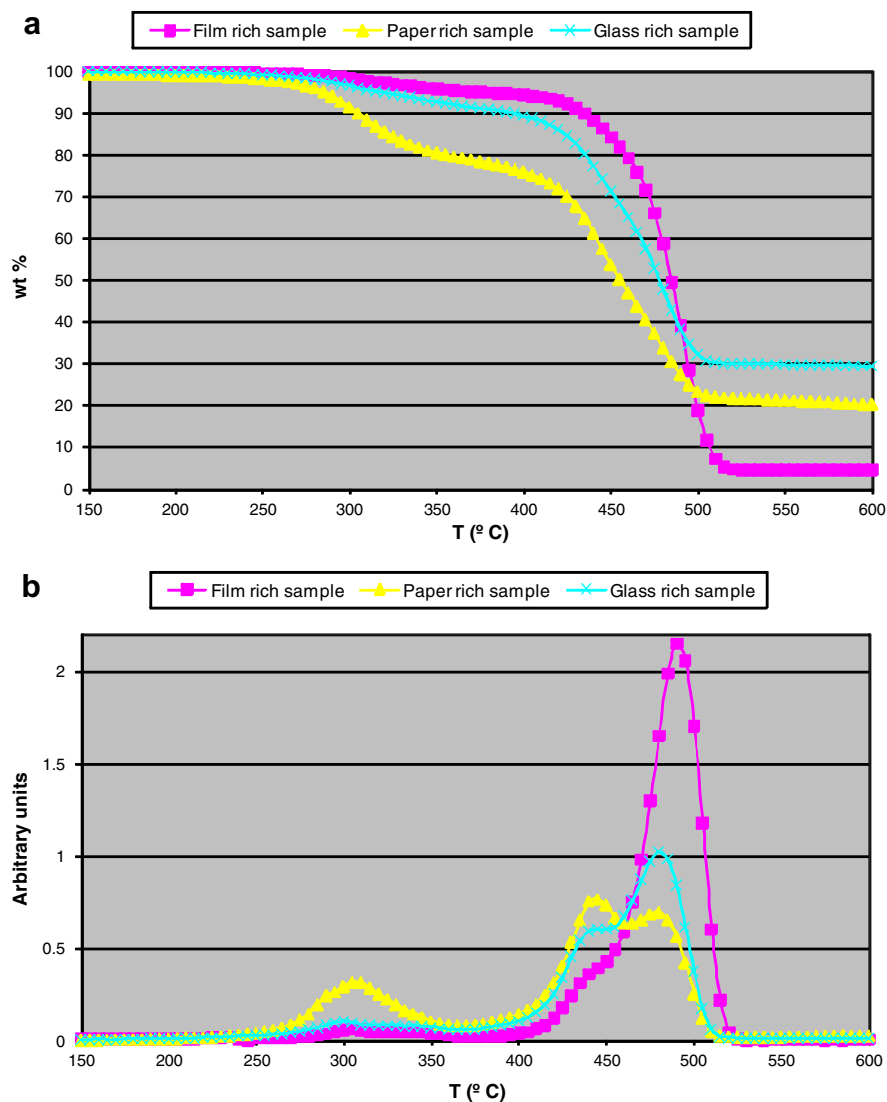


Fig. 2. TG and DTG plots of the samples pyrolysed.

**Table 4**  
Pyrolysis yields (wt.%).

| Fraction |               | Film rich sample | Paper rich sample | Glass rich sample |
|----------|---------------|------------------|-------------------|-------------------|
| Liquids  | Organic phase | 41.5             | 25.5              | 39.2              |
|          | Aqueous phase | 0.0              | 17.1              | 3.3               |
| Gases    |               | 49.9             | 29.4              | 13.8              |
| Solids   | Inorganics    | 2.1              | 11.2              | 30.8              |
|          | Char          | 6.5              | 16.9              | 12.9              |

normal release of the polymer derived vapours. Consequently, longer residence time of the vapours in the reactor could lead to higher generation of secondary and tertiary products, which most probably lead to char formation. However, no references have been found in the literature to confirm this theory.

### 3.3. Pyrolysis liquids

The results obtained in the GC–MS analyses of the organic phase of the pyrolysis liquids are presented in Table 5. For the sake of reduction, only the identified compounds with a percentage quan-

tified area greater than 1% are shown. Nevertheless, all the identified compounds have been considered in the total quantifications (aromatic hydrocarbons, alkanes, alkenes, etc.) and in the C5–C9, C10–C13 and C13–C16 fractions.

Table 5 shows that the liquids obtained in catalytic pyrolysis of packaging wastes are a mixture of hydrocarbons from C5 to C16 for the film rich sample and from C7 to C16 for the paper and glass rich samples; they have a very high content in aromatics (78–93% area) and an important proportion of products in the C5–C9 carbon atom number range. It is worth mentioning that the film rich sample yielded 78.3% area of aromatics, in spite of being composed mainly of polyolefins (80 wt.%). In the previous paper of the authors, the amount of aromatics obtained in the corresponding non-catalytic pyrolysis run was just 36.5% area (López et al., 2010). Therefore, ZSM-5 zeolite clearly promotes aromatisation reactions, which has also been reported by other authors (e.g. Vasile et al. 2001; Marcilla et al., 2009). Most of the aromatics found in the liquids derived from polyolefin rich samples pyrolysis are formed by secondary reactions. Aromatics formation is attributed to secondary reactions among the primary aliphatic products (Diels–Alder reactions and/or cyclation of olefin structures followed by dehydrogenation reactions) and these secondary reactions are favoured by catalysts as that used in this study.



**Table 5**

GC–MS analysis of the pyrolysis liquids (organic phases) (% area).

| Molecular formula                 | Assignment                | Film rich sample  | Paper rich sample | Glass rich sample |
|-----------------------------------|---------------------------|-------------------|-------------------|-------------------|
| C <sub>5</sub> H <sub>10</sub>    | Methyl-butene             | 2.0               | n.d. <sup>a</sup> | n.d. <sup>a</sup> |
| C <sub>7</sub> H <sub>8</sub>     | Toluene                   | 8.5               | 7.5               | 7.3               |
| C <sub>7</sub> H <sub>5</sub> N   | Benzonitrile              | n.d. <sup>a</sup> | 1.4               | <1.0              |
| C <sub>8</sub> H <sub>8</sub>     | Styrene                   | 29.8              | 25.3              | 40.2              |
| C <sub>8</sub> H <sub>10</sub>    | Ethyl-benzene             | 7.1               | 24.8              | 14.8              |
| C <sub>8</sub> H <sub>10</sub>    | Xylenes                   | 11.7              | 3.5               | 1.8               |
| C <sub>8</sub> H <sub>8</sub> O   | Phenyl-ethanone           | n.d. <sup>a</sup> | 1.3               | n.d. <sup>a</sup> |
| C <sub>9</sub> H <sub>8</sub>     | Indene                    | 1.0               | 1.9               | <1.0              |
| C <sub>9</sub> H <sub>10</sub>    | α-Methyl-styrene          | 3.6               | 6.0               | 6.9               |
| C <sub>9</sub> H <sub>12</sub>    | Ethylmethyl-benzene       | 3.9               | <1.0              | n.d. <sup>a</sup> |
| C <sub>9</sub> H <sub>12</sub>    | Trimethyl-benzene         | 2.8               | 1.4               | n.d. <sup>a</sup> |
| C <sub>9</sub> H <sub>18</sub>    | Dimethyl-heptene          | 1.9               | <1.0              | 3.8               |
| C <sub>9</sub> H <sub>18</sub>    | Propyl-cyclohexane        | n.d. <sup>a</sup> | n.d. <sup>a</sup> | 1.7               |
| Total C5–C9 <sup>b</sup>          |                           | 75.5              | 76.0              | 78.6              |
| C <sub>10</sub> H <sub>8</sub>    | Naphthalene               | 1.1               | 4.8               | n.d. <sup>a</sup> |
| C <sub>10</sub> H <sub>10</sub>   | Methyl-indene             | 1.2               | 2.1               | n.d. <sup>a</sup> |
| C <sub>11</sub> H <sub>10</sub>   | Methyl-naphthalene        | 2.7               | 4.0               | n.d. <sup>a</sup> |
| C <sub>11</sub> H <sub>24</sub>   | Undecane                  | 1.5               | n.d. <sup>a</sup> | n.d. <sup>a</sup> |
| C <sub>12</sub> H <sub>10</sub>   | Ethenyl-naphthalene       | n.d. <sup>a</sup> | 2.0               | n.d. <sup>a</sup> |
| C <sub>12</sub> H <sub>26</sub>   | Dodecane                  | 1.1               | n.d. <sup>a</sup> | n.d. <sup>a</sup> |
| Total C10–C13 <sup>b</sup>        |                           | 8.5               | 12.9              | 0.6               |
| C <sub>14</sub> H <sub>10</sub>   | Phenanthrene              | n.d. <sup>a</sup> | 1.4               | n.d. <sup>a</sup> |
| C <sub>14</sub> H <sub>30</sub>   | Tetradecane               | 1.0               | n.d. <sup>a</sup> | n.d. <sup>a</sup> |
| C <sub>15</sub> H <sub>14</sub>   | Phenylcyclopropyl-benzene | 1.2               | n.d. <sup>a</sup> | 1.1               |
| C <sub>15</sub> H <sub>16</sub>   | Propanediyl-benzene       | 2.0               | 2.0               | 2.7               |
| C <sub>16</sub> H <sub>12</sub>   | Phenyl-naphthalene        | n.d. <sup>a</sup> | 3.2               | 1.9               |
| C <sub>16</sub> H <sub>16</sub>   | Phenylbutenyl-benzene     | 1.6               | n.d. <sup>a</sup> | 1.4               |
| Total C13–C16 <sup>b</sup>        |                           | 5.8               | 7.3               | 7.2               |
| Total aromatic hydrocarbons       |                           | 78.3              | 92.5              | 79.8              |
| Total alkanes                     |                           | 4.5               | n.d. <sup>a</sup> | n.d. <sup>a</sup> |
| Total alkenes                     |                           | 0.8               | n.d. <sup>a</sup> | n.d. <sup>a</sup> |
| Total branched alkenes            |                           | 3.9               | 0.9               | 3.8               |
| Total naphthenes                  |                           | n.d. <sup>a</sup> | n.d. <sup>a</sup> | 1.7               |
| Total hetero-compounds (Cl, N, O) |                           | 2.2               | 2.7               | 3.2               |
| Not identified                    |                           | 10.3              | 3.9               | 11.5              |

<sup>a</sup> Not detected.<sup>b</sup> This classification includes all compounds in this range, not only those reported in the upper part of the table.

Concerning the carbon atom number fractions, the film rich sample yielded 75.5% area of C5–C9 products, while in the corresponding non-catalytic run only 35.9% area of such products and up to 46.5% area of >C13 products were obtained, even though the thermal run was carried out at higher temperatures than the catalytic one (500 and 440 °C, respectively). Even though the liquid yield in the catalytic run is lower than in the thermal run, the absolute value of C5–C9 percentage is higher than in the thermal run. Therefore, ZSM-5 zeolite is a very effective catalyst for polyolefin rich samples cracking, increasing the C5–C9 fraction which is of the most interest since such fraction resembles gasoline like fuels.

Table 5 also shows that the liquids obtained with the film rich sample contain 4.5% area of alkanes but only 0.8% area of alkenes (and none among the main components of the liquids), which are normally the main primary products generated in PE decomposition at low temperatures (Predel and Kaminsky, 2000; Ueno et al., 2010). This fact may be explained as follows; the main mechanisms proposed up to now to explain the formation of aromatics in pyrolysis of polyolefin rich samples are (1) Diels–Alder type reactions followed by dehydrogenation (Williams and Williams, 1999) and (2) direct cyclation of olefinic structures (Ji et al., 2006). Both processes, apart from contributing to olefin transformation, generate high quantities of hydrogen, which can be adsorbed in the surface of the zeolite, and so it is readily available to saturate olefins, leading to alkanes production.

The liquids obtained with the paper rich sample contain a very high proportion of aromatics (92.5% area). Such content is quite

similar to that obtained by the authors in the previous thermal experiment (96.6% area) (López et al., 2010). Table 5 shows that with this sample the C5–C9 fraction is also, as with the film rich sample, the most abundant one (76.0% area), and it is also as abundant as in the corresponding thermal run (López et al., 2010). It must not be inferred from these facts that the catalyst has almost no effect with this sample since it must be borne in mind that the thermal experiments were carried out at 500 °C while the experiments with ZSM-5 catalyst have been carried out at 440 °C, so the catalyst enables to lower 60 °C the process temperature keeping the quality of the products.

Finally, the liquids derived from the glass rich sample contained 79.8% area of aromatics, while in the thermal experiments carried out at 500 °C aromatics amounted up to 93.4% area (López et al., 2010), so it seems that with this sample the catalyst is less effective than with the other samples. A plausible explanation to this fact is that fine inorganic particles derived from brittle materials as glass, which are especially abundant in this sample, may block the zeolite pores leading to its deactivation. In addition to this, it must be also taken into account that in the previous paper (López et al., 2010) the authors concluded that the inorganic materials of this sample may have some kind of catalytic effect, and consequently there is no margin for the zeolite to highlight its effect. On the other hand, the proportion of C5–C9 hydrocarbons is the highest among the three samples (78.6% area) but it is also lower than that obtained in the previous non-catalytic runs (81.9% area) (López et al., 2010).

Table 5 shows that in every case, styrene is the most abundant product with percentage areas ranging from 25% with the paper rich sample to more than 40% with the glass rich sample. Since the PS contents of the samples is rather low (6–11 wt.%), styrene must be generated not only by direct depolymerisation of PS, but also by secondary reactions that take place in the course of pyrolysis. It is a really difficult task to propose a possible mechanism for styrene formation. First of all, it has to bear in mind that in catalytic pyrolysis of plastics both thermal and catalytic decomposition take place at the same time, so both mechanisms can be produced simultaneously. Concerning thermal decomposition, styrene is formed at a great extent from direct depolymerisation of PS but also by reactions among the primary products derived from the other polymers decomposition. On the other hand, it has been reported (Serrano et al., 2000) that styrene formation during PS catalytic decomposition with zeolites can be lower than in thermal decomposition since the previously formed styrene decomposes into other aromatic compounds as a consequence of the secondary reactions that takes place to a greater extent in presence of zeolites. This is the reason why styrene yield is usually lower in catalytic pyrolysis compared to thermal pyrolysis. As a consequence, to elucidate a possible mechanism for styrene formation in such a complex system (many difficult input materials and many more decomposition products) is not an easy task.

The second most abundant product in the liquids derived from the paper and glass rich samples is ethyl-benzene (24.8% and 14.8% area, respectively), while xylenes isomers are the second compound in abundance in the film rich sample liquids (11.7% area). The proportion of toluene in the three liquids is also significant (7.3–8.5% area). A relationship between styrene and ethylbenzene yields can be inferred from the data. For the paper and glass rich samples, which contain similar quantities of PS, the total sum of styrene and ethylbenzene is almost the same (50.1% and 55% area, respectively). In the case in which the zeolite has stronger effect (with paper rich sample), the yield of ethylbenzene is higher than that of the glass rich sample. These data could indicate that ethylbenzene is one of the products of the secondary reactions of styrene.

This composition confers pyrolysis liquids a potential use as raw chemicals source, even though it is known that the achievement of an efficient separation of such chemicals is not easy. On the other hand, although aromatics proportion is limited in commercial fuels, some oil refinery processes (e.g. catalytic reforming process) are focused on producing light aromatic hydrocarbons from naphthenes and alkanes, with the aim of obtaining high-octane reformates for gasoline blending. A highly aromatic liquid fraction coming from plastics pyrolysis may be used in such blending operations without the need of catalytic reforming processes.

The GC–MS analysis of the aqueous phases obtained in the pyrolysis of the paper and glass rich samples is shown in Table 6; this time all the identified compounds have been included. In both cases, the aqueous phase is a mixture of oxygenated compounds in water solution, being water by far the main component. The proportion of water in the aqueous phase of the paper rich sample liquid (92.4% area) is higher than that obtained in the previous work of the authors (86.1% area) (López et al., 2010), when no catalyst was used. Other research groups (Sharma and Bakhshi, 1993; French and Czernik, 2010) have reported that ZSM-5 zeolite is a good catalyst to convert oxygenated organic compounds derived from the primary decomposition of cellulose based materials into short chain hydrocarbons, which will be detected in gaseous phase. This could be the reason why there is less quantity of oxygenated compounds in the aqueous phase of the catalytic experiment compared to the thermal one. Therefore, it could be concluded that one of the roles of ZSM-5 zeolite in pyrolysis of -

**Table 6**

GC–MS analysis of the pyrolysis liquids (aqueous phases) (% area).

| Molecular formula                            | Assignment            | Paper rich sample | Glass rich sample |
|----------------------------------------------|-----------------------|-------------------|-------------------|
| H <sub>2</sub> O                             | Water                 | 92.4              | 87.0              |
| C <sub>3</sub> H <sub>6</sub> O <sub>2</sub> | Methyl acetate        | n.d. <sup>a</sup> | 0.1               |
| C <sub>3</sub> H <sub>8</sub> O              | Propanol              | 1.1               | 0.2               |
| C <sub>4</sub> H <sub>6</sub> O              | Dihydrofuran          | 0.3               | 0.2               |
| C <sub>3</sub> H <sub>6</sub> O <sub>2</sub> | Propanoic acid        | 0.5               | 0.8               |
| C <sub>5</sub> H <sub>8</sub> O              | Cyclopentanone        | n.d. <sup>a</sup> | 0.2               |
| C <sub>4</sub> H <sub>6</sub> O              | Oxolanone             | 0.7               | 1.0               |
| C <sub>5</sub> H <sub>4</sub> O <sub>2</sub> | Furfural              | 0.5               | n.d. <sup>a</sup> |
| C <sub>6</sub> H <sub>6</sub> O              | Phenol                | 0.9               | 2.0               |
| C <sub>2</sub> H <sub>4</sub>                | Butadiene             | n.d. <sup>a</sup> | 0.2               |
| C <sub>7</sub> H <sub>6</sub> O <sub>2</sub> | Benzoic acid          | 0.3               | 0.2               |
| C <sub>6</sub> H <sub>6</sub> O <sub>2</sub> | Benzenediol           | 0.2               | 0.2               |
| C <sub>6</sub> H <sub>11</sub> NO            | Caprolactam           | n.d. <sup>a</sup> | 2.0               |
| C <sub>9</sub> H <sub>10</sub> O             | Methylphenyl-ethanone | n.d. <sup>a</sup> | 0.1               |
| C <sub>8</sub> H <sub>4</sub> O <sub>3</sub> | Isobenzofurandione    | 0.4               | 0.4               |
| Total identified                             |                       | 97.3              | 94.6              |
| Total not identified                         |                       | 2.7               | 5.4               |

<sup>a</sup> Not detected.

cellulose based materials is to decrease oxygenated organic compounds generation, increasing as a consequence water production.

Since the zeolite has promoted the formation of water, in this work it has been possible to detect and quantify an aqueous phase in the liquid fraction of the glass rich sample. It has to be mentioned that in the previous work of the authors the carbon content of the glass rich sample derived liquids was 77.3 wt.%, which is quite low compared with the usual values which are obtained in plastics pyrolysis derived organic liquids (84–90 wt.%). The low carbon and high “others” content were probably produced due to the presence of an oxygenated phase (water included) dispersed in the organic liquid. It was not possible to separate such oxygenated phase from the organic phase, not even centrifuging the liquids, which indicates that it was in the form of an emulsion; other methods such as changing the ionic strength of the water phase would be needed in order to achieve such separation.

The elemental composition and higher heating value of the organic phase of the pyrolysis liquids are presented in Table 7. The results are quite in agreement with the GC–MS analyses and show that the liquid obtained from the paper rich sample has the highest carbon content (87.4 wt.%), as well as the lowest hydrogen content (10.2 wt.%), which corresponds to the great proportion of aromatics showed in Table 5. This is the reason why its H/C ratio and HHV are also the lowest ones (1.4 and 38.4 MJ kg<sup>−1</sup>, respectively). On the contrary, the H/C ratio and HHV of the liquid derived from the film rich sample are the highest ones (1.8 and 44.4 MJ kg<sup>−1</sup>, respectively), corresponding to the higher proportion of non-aromatic compounds in its composition. It is worth mentioning that the three liquids have very high HHV, similar to those of conventional liquid fuels, so they can be considered as an appropriate

**Table 7**Elemental composition (wt.%) and HHV (MJ kg<sup>−1</sup>) of the organic pyrolysis liquids.

| Sample              | Film rich sample | Paper rich sample | Glass rich sample |
|---------------------|------------------|-------------------|-------------------|
| C                   | 84.6             | 87.4              | 84.7              |
| H                   | 12.9             | 10.2              | 11.5              |
| N                   | <0.1             | 0.1               | 0.2               |
| Cl                  | 0.3              | 0.9               | 0.2               |
| Others <sup>a</sup> | 2.3              | 1.5               | 3.4               |
| H/C ratio           | 1.8              | 1.4               | 1.6               |
| HHV                 | 44.4             | 38.4              | 41.9              |

<sup>a</sup> By difference.

alternative to fossil fuels. Table 5 also shows that some of the chlorine coming from PVC ends in the liquid fraction; the amount of chlorine is especially high in the liquid derived from the paper rich sample (0.9 wt.%), which is the consequence of the high chlorine content of this sample. The presence of chlorine is a handicap to make use of pyrolysis liquids. A method for preventing transferring chlorine to pyrolysis oils is presented in Section 3.6.

The results in this section have shown that ZSM-5 zeolite has a strong effect in the pyrolysis liquids. However, the quantity and cost of the catalyst is a key factor in the economy of the process. The authors have studied at the same time of this work (unpublished yet) the regeneration capacity of ZSM-5 zeolite after plastic wastes pyrolysis process. The results obtained in such study have proved the possibility of zeolite regeneration, leading to the same results as those obtained in the pyrolysis runs carried out with fresh ZSM-5 zeolite. These results make ZSM-5 zeolite be a very interesting catalyst for plastic wastes pyrolysis.

### 3.4. Pyrolysis gases

Table 8 shows that pyrolysis gases are composed of hydrocarbons ranging from C1 to C6, hydrogen and some carbon dioxide and monoxide. In all the cases, C3 and C4 are the main components of the gases. In the non-catalytic pyrolysis runs carried out by the authors in the previous work (López et al., 2010), noticeable lower proportions of C3 and C4 compounds were obtained. Therefore, ZSM-5 zeolite promotes C3–C4 fractions formation. Wei et al. (2010) and Huang et al. (2010) have also reported that C3 and C4 fractions are the main gaseous components in ZSM-5 catalytic pyrolysis of PE and PP rich samples.

The paper rich sample generated high quantities of CO and CO<sub>2</sub>, which was also reported in the first part of this work (López et al., 2010) and it is most probably due to its high paper content, since cellulose is a high oxygen containing polymer. It has to be mentioned that the proportions of CO and CO<sub>2</sub> obtained in this work (8.7 and 20.4 wt.%, respectively) are lower than those obtained in the non-catalytic experiments in the previous paper (11.6 and 33.1 wt.%, respectively). This may be because some of the oxygen has been used to form water which condensed in the aqueous liquid fraction, which, as it has been mentioned before, is more abundant in the catalytic than in the non-catalytic experiment. It is also worth noting that the paper rich sample generated the highest quantity of H<sub>2</sub> in the gases, probably due to the fact that the liquids of this sample are the ones that contain the highest quantity of aromatics, and aromatisation reactions are normally accompanied by H<sub>2</sub> production.

The glass rich sample showed behaviour halfway between those of the other two samples; it generated more quantity of C3–C4 fraction than the paper rich sample and less than the film rich sample, due to its proportions of PE and PP, which are halfway between those of the other samples. Similarly, it also generated less

CO and CO<sub>2</sub> than the paper rich sample and less than the film rich sample, probably due to the relatively high proportion of cellulose based materials that this sample contains.

The HHV of the gases depends on the CO<sub>2</sub> content and consequently the paper rich sample, which is the one with the highest content of CO<sub>2</sub>, has the lowest HHV (36.7 MJ kg<sup>-1</sup>). Anyway, the HHV of any of the pyrolysis gases is quite high, close than that of natural gas (48–53 MJ kg<sup>-1</sup>). For this reason pyrolysis gases may be used as energetic source for the process.

### 3.5. Pyrolysis solids

The composition of pyrolysis solids is presented in Table 9. Given that it was not possible to separate the catalyst from the pyrolysis solid products, the latter were characterised mixed with the catalyst. The catalyst and the inorganic material coming from the original samples are the reason why the “ash” classification is the highest one in all the cases (56–76 wt.%), especially in the solid of the glass rich sample. Comparing to the results obtained in the previous paper of the authors in experiments without catalyst (López et al., 2010), a much higher H/C ratio has now been obtained (1–1.6 now and 0.5 then). This is probably due to the fact that the thermal runs were carried out at 500 °C, while the catalytic runs were performed at 440 °C; at this lower temperature small quantities of the raw material may have not been decomposed.

The film rich sample derived solid is the one with the highest H/C ratio, which may indicate that more unconverted plastic matter is present in this solid than in the other ones. Finally, it must to be mentioned that the pyrolysis solids have a significant chlorine content (3.7–5.5 wt.%), the highest one that obtained in the solid coming from the paper rich sample, which is in accordance with some published results which report that cellulose based materials are capable of fixing the chlorine in the solid fraction of the pyrolysis process (Fontana et al., 2000; Kuramochi et al., 2008).

The pyrolysis solids may look at first sight as a useless product of the pyrolysis process, which most probably would have to be landfilled. At best, the inorganic material and the catalyst could be recovered and the organic fraction could be used for energy recovery. Bernardo et al. (2009, 2010) have recently published some works warning about the potential ecotoxicology of pyrolysis derived chars due to their content in heavy metals and organic contaminants. So whatever the final use of these solids is, a pre-treatment step aimed to improving the material quality would probably be necessary prior to their final disposal.

### 3.6. Avoiding water and chlorine in pyrolysis liquids by means of stepwise pyrolysis

Stepwise pyrolysis is an alternative to avoid transferring the chlorine that comes from PVC containing plastic mixtures to

**Table 8**  
GC–TCD/FID analysis (wt.%) and GVC (MJ kg<sup>-1</sup>) of pyrolysis gases.

| Sample          | Film rich sample | Paper rich sample | Glass rich sample |
|-----------------|------------------|-------------------|-------------------|
| H <sub>2</sub>  | 0.5              | 1.2               | 0.7               |
| CO              | 0.6              | 8.7               | 3.7               |
| CO <sub>2</sub> | 1.4              | 20.4              | 11.7              |
| Methane         | 1.4              | 11.0              | 4.3               |
| Ethane          | 1.4              | 4.9               | 4.9               |
| Ethene          | 5.1              | 12.5              | 5.8               |
| C3              | 29.6             | 24.3              | 27.3              |
| C4              | 32.0             | 15.2              | 25.2              |
| C5              | 19.8             | 1.8               | 15.1              |
| C6              | 8.3              | <0.1              | 1.2               |
| HHV             | 48.3             | 36.7              | 42.6              |

**Table 9**  
Moisture, elemental composition in wet basis (wt.%) and HHV (MJ kg<sup>-1</sup>) of pyrolysis solids.

| Sample              | Film rich sample | Paper rich sample | Glass rich sample |
|---------------------|------------------|-------------------|-------------------|
| Moisture            | 5.0              | 6.5               | 2.2               |
| C                   | 26.8             | 27.4              | 14.5              |
| H                   | 3.6              | 2.4               | 1.2               |
| N                   | 0.5              | 0.6               | 0.8               |
| Cl                  | 3.7              | 5.5               | 4.0               |
| Ash                 | 57.7             | 56.2              | 75.9              |
| Others <sup>a</sup> | 2.7              | 1.4               | 1.4               |
| H/C ratio           | 1.6              | 1.1               | 1.0               |
| HHV                 | 14.5             | 10.4              | 11.0              |

<sup>a</sup> By difference.



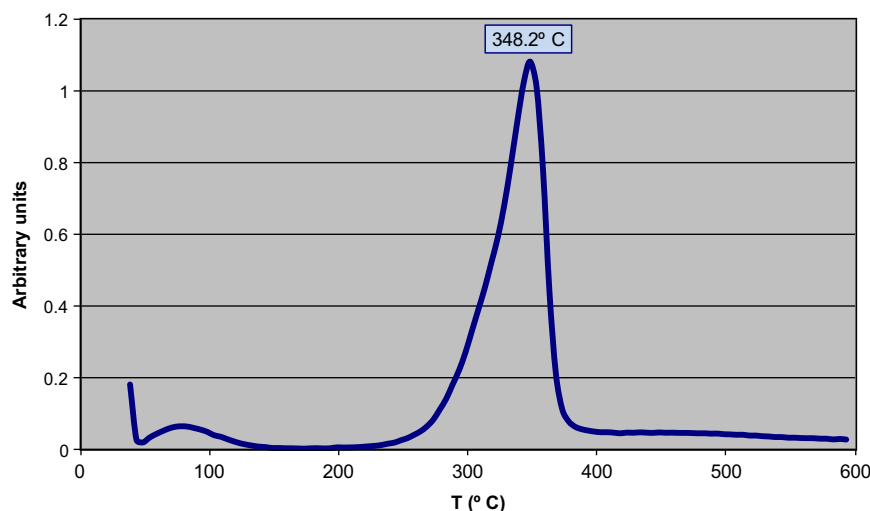


Fig. 3. DTG plot of a sample of magazine paper.

pyrolysis liquids. It consists in a previous low temperature step in which the PVC is partially decomposed generating HCl, which is removed from the reaction medium and therefore cannot react with the pyrolysis organic vapours (Hornung et al., 1999; Sakata et al., 2004). In a previous paper (López et al., 2011), the authors obtained quite good ratios of dechlorination of the liquids of a PVC containing plastic mixture by means of a 300 °C–60 min step previous to pyrolysis.

As it has been discussed above, the presence of paper in packaging waste samples leads to the generation of an aqueous phase in the liquid fraction; this involves that a separation process will be required to remove water after pyrolysis. In order to know the behaviour of paper under pyrolysis conditions, a paper sample coming from magazines was analysed by thermogravimetric analysis. The first derivative plot of the weight loss of the sample as a function of temperature is shown in Fig. 3. It can be seen that the maximum degradation rate of paper takes place at around 350 °C, which is quite close to the temperature at which the maximum degradation rate of PVC first decomposition takes place ( $\approx 300$  °C), and far enough from the temperature at which the maximum degradation rates of the main packaging plastics takes place ( $\approx 425$  °C for PS,  $\approx 450$  °C for PET,  $\approx 475$  °C for PP and  $\approx 485$  °C for PE), as it has been proved by the authors and other research groups (e.g. Bhaskar et al., 2005). Additionally, Korkmaz et al. (2009) have reported in a study about pyrolysis of tetrapack that the temperature at which the maximum degradation rate of the cardboard takes place is about 360 °C. Therefore, paper-like materials which are usually found in packaging wastes decomposed in a relatively low temperatures range.

A stepwise pyrolysis experiment with a previous step at 300 °C during 60 min was carried out with the paper rich sample in order to know if it is possible to remove water previous to the pyrolysis process itself. Since the objective of this experiment was just water removal, it was carried out without catalyst and at 500 °C, which are the same conditions as those used in the previous paper of the authors (López et al., 2011). The pyrolysis yields and the characteristics of the liquids are presented in Tables 10 and 11.

Table 10 shows that complete removal of water is achieved if stepwise pyrolysis is applied. As expected, the organic liquid yield was higher and the gas yield lower than those obtained in the catalytic experiment (Table 4), while char yields were very similar (16.0 wt.% in stepwise and 16.7 wt.% in the catalytic experiment). It has to be mentioned that the aqueous phase yield obtained in stepwise pyrolysis was lower (16.7 wt.%) than in the case of cata-

Table 10

Pyrolysis yields obtained in stepwise pyrolysis with paper rich sample (wt.%).

| Fraction |               | Previous step (300 °C) | Pyrolysis step (500 °C) |
|----------|---------------|------------------------|-------------------------|
| Liquids  | Organic phase | 0.0                    | 35.0                    |
|          | Aqueous phase | 16.7                   | 0.0                     |
| Gases    |               | 20.9                   |                         |
| Solids   | Inorganics    | 11.4                   |                         |
|          | Char          | 16.0                   |                         |

Table 11

Elemental composition (wt.%) and HHV ( $\text{MJ kg}^{-1}$ ) of the organic pyrolysis liquids obtained in stepwise pyrolysis of the paper rich sample.

| Sample              | Paper rich sample |
|---------------------|-------------------|
| C                   | 60.9              |
| H                   | 10.2              |
| N                   | 0.2               |
| Cl                  | 0.2               |
| Others <sup>a</sup> | 28.7              |
| H/C ratio           | 1.4               |
| HHV                 | 41.5              |

<sup>a</sup> By difference.

lytic pyrolysis (19.2 wt.%), which can be explained as follows. Table 11 shows that the carbon content of the organic liquid obtained in stepwise pyrolysis is quite lower than that obtained in the catalytic experiment (60.9 and 87.4 wt.%, respectively), which produces a great “others” content in the former, probably due to high proportions of oxygen solved in the organic fraction. Although this high oxygen content could be a limitation for liquid fraction applications, quite a similar elemental composition can be found in the bio-oils coming from biomass pyrolysis (Demirbas, 2007; Bhat-tacharya et al., 2009), for which a wide range of applications and purification processes are being developed nowadays (Mohan et al., 2006).

Additionally, the low temperature previous step produced a high ratio of dechlorination of the sample, leading to an organic liquid fraction with only 0.2 wt.% of chlorine instead of the 0.9 wt.% which was obtained in the conventional catalytic pyrolysis run with this same sample (Table 7). This is a promising result since almost 80 wt.% of dechlorination has been achieved. Therefore, a

previous low temperature step is an effective way of avoiding both chlorine and water in the pyrolysis liquids, when PVC and/or paper containing packaging wastes are pyrolysed.

#### 4. Conclusions

The results presented in this paper have been obtained at lab scale with three plastics-rich residues coming from a waste packaging sorting industrial plant. Since the composition of the samples is intrinsically fluctuating and the installation used is not an industrial one, the results must be considered as an interesting approach for developing new plastic waste management solutions. Pyrolysis is an attractive alternative for chemical recycling of complex plastic wastes and the use of catalyst can significantly enhance the process.

The main conclusions that can be drawn from the results of this work are the following. ZSM-5 zeolite is a suitable catalyst for obtaining C3–C4 rich gases and liquids with high aromatic content at relatively low temperatures ( $\approx 440^\circ\text{C}$ ) even if the raw material contains great quantities of polyolefins. However, when there are high proportions of inorganic components in the raw materials, physical blockage and subsequent loss of the catalyst activity takes place, and no effect in the pyrolysis products is observed. In such cases, a thermal process at higher temperatures would probably be a more effective way of obtaining valuable products. When the waste sample contains significant quantities of paper and cellulose based materials, the formation of a two-phase liquid fraction can be avoided by means of a previous low temperature step ( $\approx 300^\circ\text{C}$ ), in which the oxygen rich aqueous phase is removed. In these cases, an organic liquid which resembles bio-oils coming from biomass pyrolysis is obtained, which contains less carbon and more oxygen than the usual in pyrolysis liquids. ZSM-5 zeolite promotes the formation of water, so it should be added to the process after the separation of the aqueous phase in the previous low temperature step. In any case, stepwise pyrolysis is recommended to reduce the chlorine content in the liquid fraction when PVC is one of the raw materials.

The pyrolysis liquids and gases obtained in pyrolysis of packaging wastes are high HHV products which can be used as alternative fuels for heat and power generation. Additionally, the liquid fraction can be considered a source of valuable chemicals, since styrene, xylenes, toluene or ethyl-benzene are obtained in high proportions. Finally, metals and glass of the solid fraction may be recovered for recycling processes.

This and the previous paper of the authors, both devoted to analyse the behaviour of different real packaging samples in pyrolysis, have shown that such process is a promising technique for feedstock recycling of packaging plastic wastes. It is a versatile process which can be adapted to the nature and composition of the raw material in order to make the most of the pyrolysis products. Variations in the process, such as the use of catalyst and carrying out the process in more than one step, enable to produce good quality fuels and chemicals from poor-valued mixed packaging wastes.

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